جامعة أبوظبي
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QUAD TILT WING WIND TESTING

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Introduction:

UAV's or better known as Unmanned Air Vehicle is a recent field where there has been many significant research and developments towards it. There have been many uses of UAV; for example, they have provided a huge amount of help in surveillances where humans cannot go. Furthermore, they are many types of UAV's and they are listed as:

1. Fixed Wing UAV: These UAV able to fly for longer range for longer time. However, they need a runaway and landing gear equipment.
2. Rotary Wing UAV: These UAV maneuver much better than Fixed Wing UAV, these are also used in surveillance as well as SAR operations.
3. Hybrid Designs UAV: These UAV combine both the Fixed Wing and Rotary Wing UAV's motor and thus, able to hover and use thrust motors for better movements. Moreover, this produces more complexity and mechanical problems when making this UAV.

Problem Statement:

Is to design an UAV for indoor and outdoor surveillance that has also the ability to takeoff vertically and as able to over horizontally for a longer period.

Objective:

To design and test the characterization of the quad-tilt wing UAV through tunnel testing. Validating the design through experimental data for similar platforms.

Methodology:

In this journal, the paper has been presented into three sections:

1. Aerodynamics Design of the SUAV
2. Wind tunnel experiment and Ansys simulation.
3. Comparison of Simulation and Experimental test results

Aerodynamics design of SUAV:

The journal explains the conceptual designs of quad tilt wing UAV and furthermore, explains how it is simple for this UAV to takeoff vertically and have tilt wings for horizontal motion. As the journal explains the concept of how rudder at the tail, motor and wings to form an angle of attack are utilized in making the UAV move. The pervious sentences also corelates with the basic information on how rudders, motors, and the angle of attack plays a vital role in producing drag and lift for the plane.

Furthermore, the journal goes into depth detail of selections process of the propellers and motors. Therefore, high shaft with high torque generation were needed. This is due to the fact, once the UAV can takeoff at 180W, but it decreases significantly when it comes to horizontal movement which is 58W. When it came down to selection of the motors, the journal was able to suggest Silver Series 35 motor driver, and this was done in the simulation of MotoCalc and catalogs.

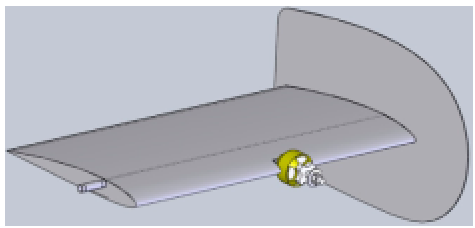

FIGURE 1 DESIGN OF THE WING

To find the required selection of propellers, the people in the journal were able to create their own thrust bench was designed. Their criteria were based on the diameter and pitch of the propeller and were tested from the size of 27.5 x 20 to 40 x 12.5cm electric motors. The following result was produced from the test run on the thrust bench.

Table 3. Maximum Thrust Test Results

Prop Size (cm)	Current (A)	Thrust (g)	Thrust/current (g/A)	Power (W)
27.5 × 20	25.3	1,260	49.8	280.8
30 × 20	31.5	1,577	50.1	349.6
33 × 16.5	32.0	1,812	56.6	355.2
33 × 20	32.0	1,690	52.8	355.2
35 × 17.5	32.0	1,876	58.6	355.2
35 × 21.2	32.0	1,728	54.0	355.2

Table 4. Thrust Test Results for Nominal Hover Flight Thrust

Prop Size (cm)	Current (A)	Thrust/current (g/A)	Power (W)
27.5 × 20	21.4	52.6	237.5
30 × 20	18.4	61.1	204.2
33 × 16.5	14.6	77.0	162.0
33 × 20	16.3	69.0	180.9
35 × 17.5	13.9	80.9	154.3
35 × 21.2	15.6	72.1	173.2

FIGURE 2 PROPELLER TESTING

The result was a propeller of the size of 35 x 17.5cm as it can produce sufficient current for the thrust at 60 km/h airspeed. This helps in controlling the plane in y-axes or more commonly known as yaw. The yaw is controlled from the vertical flight mode and can balance with high angle of attack.

Now we come to fuselage and wing design, this is the part where Cd (Coefficient of Drag) plays a major role on the usage of fuel. The main role was to reduce the Air Drag coefficient while generating an optimizing lift. This will make the most lightweight structure and less fuel cost/battery cost. The wing shape was chosen as "rectangular prism with rounded nose section" when viewed from the tip it shows "symmetrical wing with long straight sides". Now the importance of having a symmetrical wing is that it reduces the turbulent produced from the front and the back as well as the concept of straight sides reduces the additional losses. Going into in depth of straight sides is that it has 0-degree wind angle. This reduces the induced drag and vortices which are usually produced when the wing is about to lift. The journal proceeds with NACA 4412 wing profile, as a starting point for the wing design. We know that from aerodynamics is that, this airfoil consists of 12% thickness and 4% camber line. When it comes to Newton law of third motion, every action has an opposite reaction, the wing was able to pull back the UAV with backward force of 3.2N.

Simulation and Testing

Ansys Simulation:

In this section, the journal explains the most significant parts of the UAV tilt wing design that concerns the moment, lift and drag forces. In addition, the performances of vortices that decreases after the wing tips. They were able to perform Ansys Airfoil simulation regarding various angles of attack of different NACA airfoils. The table shows the test on different NACA airfoils

Table 6. Angle of Attack and Drag Values for Several Wing Configurations to Generate 1.125 kg Lift at 40 km/h

Wing configuration	Angle of attack (degrees)	Drag force (N)
NACA 4412 with 25-cm chord length	7.9	3.11
NACA 4412 with 25-cm chord length and large winglet	5.6	2.84
NACA 4416 with 25-cm chord length	7.6	3.53
NACA 4416 with 25-cm chord length and large winglet	5.5	3.36
NACA 4420 with 25-cm chord length	7.5	3.09
NACA 4420 with 25-cm chord length and large winglet	5.4	3.05
NACA 4416 with 20-cm chord length	10.6	3.09
NACA 4416 with 20-cm chord length and large winglet	8.0	3.02
NACA 4420 with 20-cm chord length	10.3	3.15
NACA 4420 with 20-cm chord length and large winglet	7.8	3.08

FIGURE 3 NACA AIRFOILS FOR DIFFERENT ANGLE OF ATTACK

The results that produced was obvious, as we learned from that having a longer wingspan and long chord length was able to produce more lift and drag ratio. As we know that, if we do have a thinner and small cambered line, this causes the separation to happen at a lower angle of attack causing a stall. So longer wingspan makes it to have air separation which results in stall at higher angle of attack. As the wings of SUAV were submerged completely in stream, NACA 4412 was producing more lift with additional drag that results in loss of speed. Therefore, NACA 2410 was selected as it showed up to angle of 20-degree which it was able to avoid spans wise flow at the wing roots and after that faces turbulence.

Here is the thing, from the ANSYS simulation we can figure out the downwash that is affecting the wing. From the results, it showed that the rear side of the wings were negatively affected by production from the front wings.

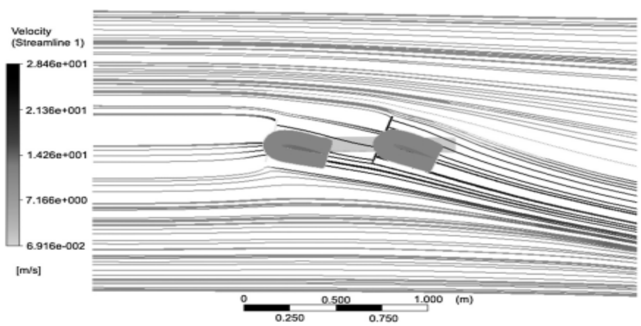

Figure 4 DOWNWASH AT THE REAR WING SIDE

To reduce the problem of downwash, the front and rear wing were located at the same vertical level which also gives a rear angle a higher angle of attack.

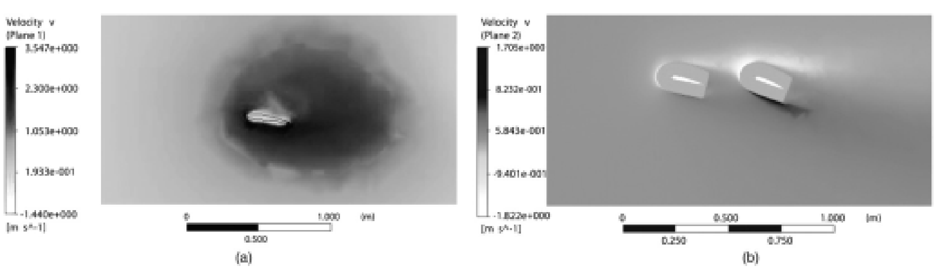

Figure 5 SPAN WISE AIRFLOW ON WINGS WITHOUT AND WITH WINGLETS

Wind Tunnel Testing

Using a half model to test the results of the flow tips values as well as the values of forces of drag and lift. It was conducted in to two stages where, the lift and the drag measurements were taken on the wing without any motor revolutions involved. This resulted in the production of values such as strength of the wing, wing-tilted servo to hold the wings at a desired angle and evaluation of wing profile. The second measurement was at full range of the flight speeds, concerning every single measurement to be taken such as pitching moment measurement to get the optimized wing tip shape, required motor, and wing incidence angles.

FIGURE 6 WIND TUNNEL TESTING ON HALF BODY WING

The equations that were used in the calculation of the lift and drag are as follows:

$$F_l = \frac{1}{2}\rho V^2 S C_l$$

$$F_d = \frac{1}{2}\rho V^2 S C_D$$

Downwash calculations included are:

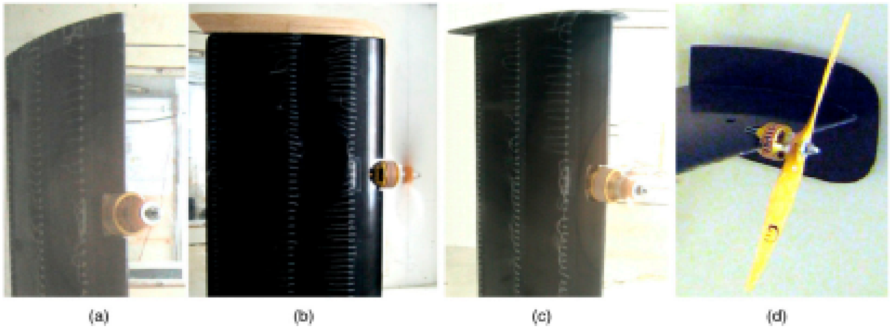

Figure 7 Different types of testing on wings

The testing was done on the complete half model, was the winglet shape, nominal flight conditions, pitching moment regulations and compensation of battery variations. The accuracy on these testing aspect is that for wing tip designs they were able to test varieties of different wingtip designs for a low aspect ratio. The results showed a moderate-size plate is used as it prevented the span wise flows and size is a small end plate thus making the SUAV light.

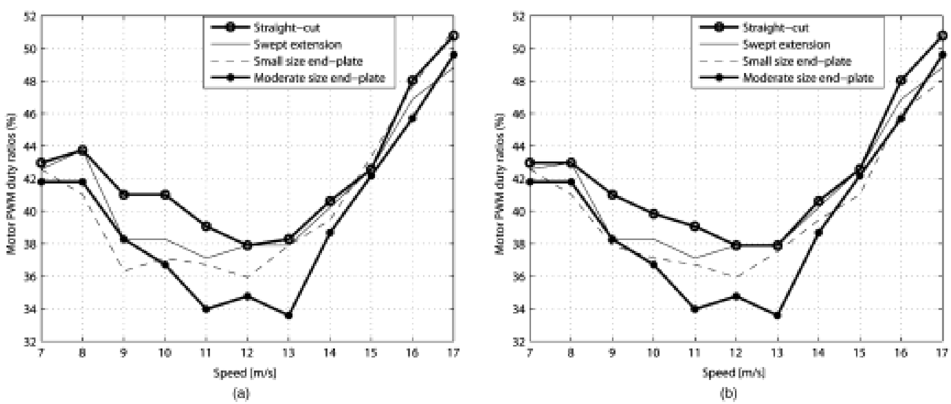

Figure 8 Motor power against speed

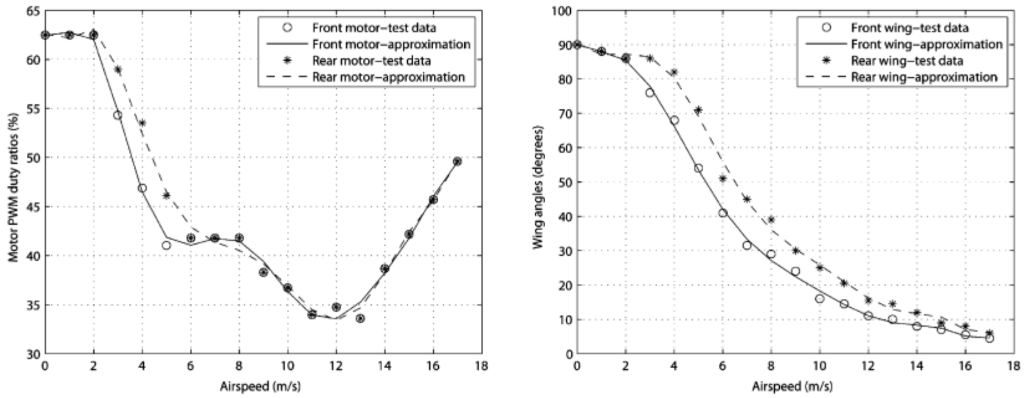

FIGURE 9 MOTOR AGAINST AIRSPEED

In nominal flight characterization, a propulsion system was established from front and rear motor PWM as we can see from the graph, as the speed increases PWM values increases to match the increased drag on the vehicle. But for the rear side of motor, there is low speed. However, the angle at the rear motor side is higher due to the cause of downwash which increases the front wing by 15-degrees.

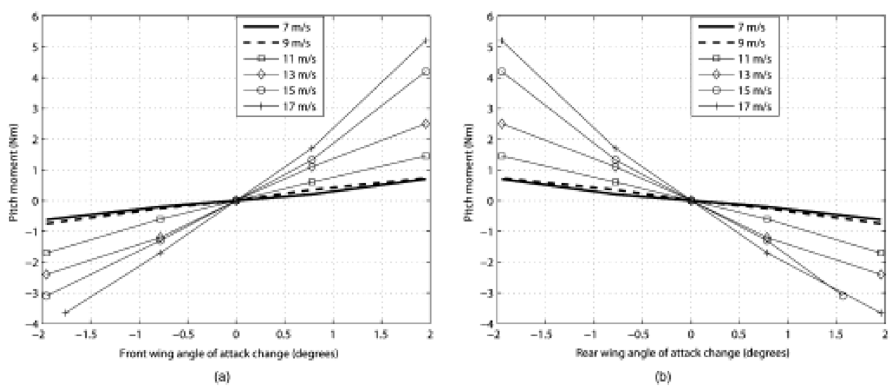

FIGURE 10 PITCHING MOMENT AGAINST ANGLE OF ATTACK

When it came to Pitching moment, the journal explains us that they were two conductions of test regarding the half model were adjusted to steady flight conditions so that there could be testing of wing incidence angle against motor power. The second consisted the same flow conditions except the front wing incidence was increased whereas, the rear wing incidence was decreased. The result was remarkable as the airspeed increases, the lift decreases as of low incidence angle. This causes the pitch moment to stabilize the vehicle with just 0.5-N.m.

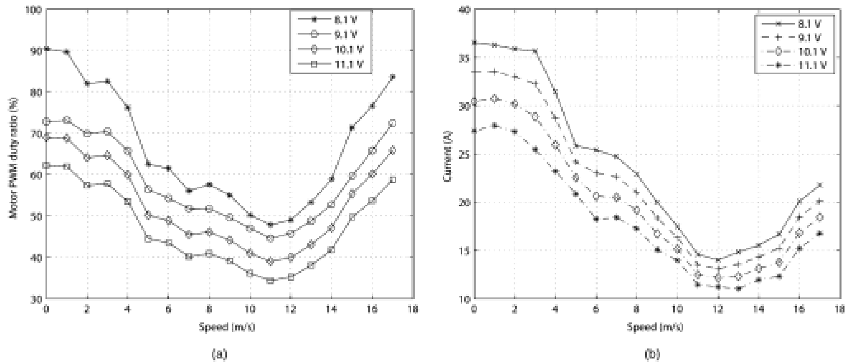

FIGURE 11 BATTERY CONSUMPTION AGAINST SPEED

Moreover, the journal explains the flight loss due to loss of motor power. They were able to conduct the same conditions for a different speed variation. The result showed that the power consumed is small and with that the mean values of the motor PWM's were used to get the current of the motors.

Comparison of Simulation and Testing:

The journal explains at the end, when doing Ansys simulation it was seen that the rear wings were going to be affected by the downwash effect. Due to the simulation, they were able put the rear wing at higher angle which is 2 and 1.4 – degrees higher than front wings. During experimentation they were bale to get the angle to be 6 and 1.5-degrees respectively.

In addition, there was simulation of span wise air flow which were being generated at the wing tips, which was why the winglets were introduced. Therefore, the winglets introduced the long chord lengths wings which were more efficient in reducing the drag forces. This simulation was validated through the testing where four different types of wing shapes.

Conclusion:

In the end, the journal was able to illustrate the SUAV into many sectors such as the vertical take off and movement across the horizontal movements. The summarization of the report is given such as that brushless motor and 35 x 17.5 cm size propellers were used. Furthermore, a fuselage design was made to reduce the drag coefficient as much as possible. NACA 2410 airfoil was used, due to experimentation and capability of it generating lift. The lift and the drag coefficients were determined at the end and they were able to figure the optimum wing size. Rear and front motor PWM were used and testing of the motors against angle of attack and speed were given in the graph. (Cetinsoy, Hancer, Oner, Sirimoglu, & Unel, 2012)

However, in this report we saw that the impact of this environment is like other platforms are able to produce more of flexibility in the terms of flight conditions. Future recommendation does state that they are going to fly the vehicle autonomously at a given trajectory. In this journal we did come across terminologies that were covered such as downwash, the axes of plane, vortices, stall etc. Moreover, the important aspect of the journal that places an important role is how they were able to make their own wind tunnel and do the testing an in top of that, verifying their own simulation results. To conclude this report, the journal illustrates all the objectives set by our mentor and this opens a new pathway of understanding aerodynamics.

Bibliography

Cetinsoy, E., Hancer, C., Oner, K. T., Sirimoglu, E., & Unel, a. M. (2012). Aerodynamic Design and Characterization of a Quad Tilt wing testing. *Quad tilt wing testing via wind tunnel*, 14.